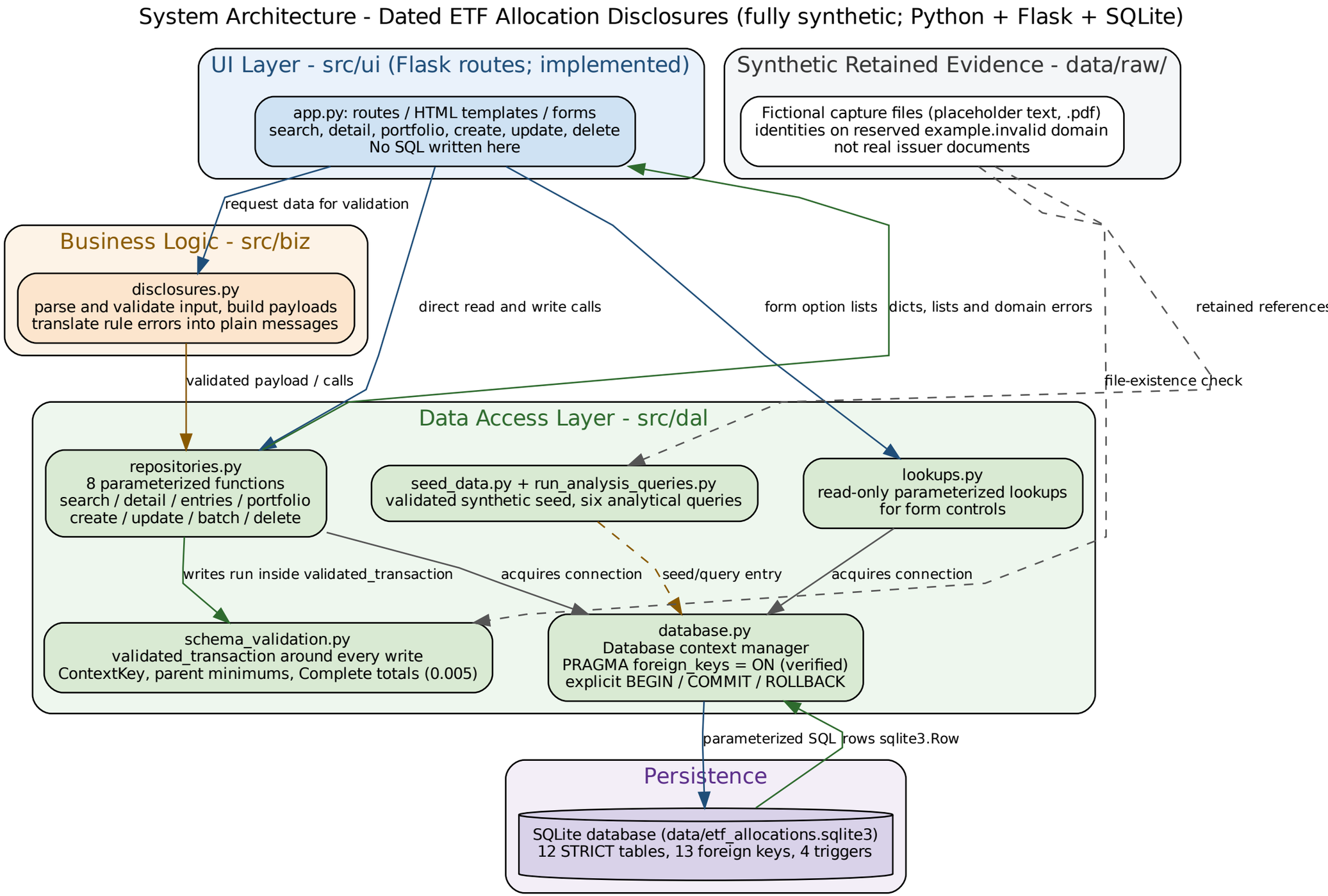
System Architecture - Dated ETF Allocation Disclosures (fully synthetic; Python + Flask + SQLite)
UI Layer - src/ui (Flask routes; implemented)
app.py: routes / HTML templates / forms
search, detail, portfolio, create, update, delete
No SQL written here
Synthetic Retained Evidence - data/raw/
Fictional capture files (placeholder text, .pdf)
identities on reserved example.invalid domain
not real issuer documents
request data for validation
Business Logic - src/biz
disclosures.py
parse and validate input, build payloads
translate rule errors into plain messages
direct read and write calls
form option lists
dicts, lists and domain errors
retained references
validated payload / calls
file-existence check
Data Access Layer - src/dal
repositories.py
8 parameterized functions
search / detail / entries / portfolio
create / update / batch / delete
seed_data.py + run_analysis_queries.py
validated synthetic seed, six analytical queries
lookups.py
read-only parameterized lookups
for form controls
writes run inside validated_transaction
acquires connection
seed/query entry
acquires connection
schema_validation.py
validated_transaction around every write
ContextKey, parent minimums, Complete totals (0.005)
database.py
Database context manager
PRAGMA foreign_keys = ON (verified)
explicit BEGIN / COMMIT / ROLLBACK
parameterized SQL rows sqlite3.Row
Persistence
SQLite database (data/etf_allocations.sqlite3)
12 STRICT tables, 13 foreign keys, 4 triggers